

THE SENTINEL HYBRID-SOC

SOC Architecture Reference Diagram

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1. Architecture Overview

The Sentinel Hybrid-SOC is a three-layer security architecture combining an on-premises endpoint monitoring layer, a cloud telemetry layer, and an automated response layer. All components feed into a centralized Wazuh SIEM dashboard — providing a single pane of glass across local and cloud infrastructure.

Layer 1 — Endpoint	Layer 2 — Cloud	Layer 3 — Automation
Mac M4 host running Wazuh Manager via Docker. Kali Linux ARM64 VM (UTM) as the monitored agent and attack platform.	AWS CloudTrail logs all API calls to S3. Wazuh pulls them every 5 minutes via the aws-s3 wodle. GuardDuty provides additional cloud threat detection.	Shuffle SOAR receives Wazuh webhook alerts and executes automated containment workflows — isolating the Kali endpoint in under 60 seconds.

2. Local Network — 10.11.3.0/22

Device	IP Address	Role	Key Ports	Criticality
Lab Gateway	10.11.0.1	Network boundary — DNS, HTTP, DB2-Admin	53, 80/8080, 6789, UDP 5353	Critical
Mac M4 Host	10.11.3.152	Wazuh Manager + Dashboard + Indexer via Docker	443, 55000, 1514, 9200	Critical
Kali Linux (UTM)	10.11.3.106	Wazuh Agent + attack platform + target endpoint	22, 1514 (outbound)	High

3. Wazuh Docker Stack — Mac M4 Host

Container	Image	Port(s)	Role
single-node-wazuh.manager-1	wazuh/wazuh-manager:4.14.4	1514, 1515, 55000	Core SIEM — rule engine, log analysis, active response
single-node-wazuh.dashboard-1	wazuh/wazuh-dashboard:4.14.4	443 → 5601	Web UI — alerts, threat hunting, MITRE dashboard
single-node-wazuh.indexer-1	wazuh/wazuh-indexer:4.14.4	9200	OpenSearch — stores and indexes all alert data

4. AWS Cloud Infrastructure — us-east-1

Component	Name / ID	Role
CloudTrail	sentinel-trail (multi-region)	Logs every API call in the AWS account → stores to S3
S3 Bucket	aws-cloudtrail-logs-573469083840-2757da7f	Stores CloudTrail JSON logs — versioning enabled
IAM User	wazuh-cloudtrail-reader	Simulated stolen credentials — S3ReadOnly, no Secrets Manager perms
Secrets Manager	sentinel-test-secret (us-east-1)	Attack target — GetSecretValue calls blocked by IAM policy
GuardDuty	Active in us-east-1	Additional threat detection — anomaly and behavior analysis
aws-s3 wodle	Wazuh integration module	Polls S3 every 5 minutes — ingests CloudTrail logs into Wazuh

5. Data Flows — How Signals Travel

Flow	Path	Protocol / Method	Latency
Endpoint → Wazuh	Kali Agent → Wazuh Manager	AES-256 / TCP 1514	Real-time
Wazuh → Indexer	Manager → Indexer (OpenSearch)	HTTPS / TCP 9200	Real-time
Wazuh → Dashboard	Indexer → Dashboard	HTTPS / TCP 443	Real-time
Wazuh → Shuffle	Manager → Shuffle (Level 12+ alert)	Webhook HTTP POST	< 2 seconds
Shuffle → Kali (isolation)	Shuffle → Wazuh Active Response → Kali agent	iptables command push	< 60 seconds
CloudTrail → S3	AWS API Gateway → S3 bucket	AWS internal	Seconds
S3 → Wazuh (wodle)	aws-s3 wodle polls S3 → Manager ingests logs	HTTPS / IAM credentials	Every 5 min
Attack → CloudTrail	Mac CLI → AWS API → AccessDenied → CloudTrail logs	HTTPS / AWS CLI	Milliseconds

6. Attack Simulations — Detection Results

Attack	Tool	MITRE Tactic / Technique	Wazuh Rule	Result
SSH Brute Force	SSH shell loop	TA0006 · T1110 · T1110.001	Rule 5712 (Lv 10)	✓ Detected
File Modification	Atomic Red Team	TA0040 · T1486	Rule 550/554	✓ Detected
Cloud Credential Theft	AWS CLI	TA0001·TA0006·TA0007 · T1526·T1528	Rule 80250 (Lv 5)	✓ Detected

7. Security Protocols in Use

Protocol	OSI Layer	Where Used	What It Protects
AES-256	Layer 6	Wazuh Agent → Manager (TCP 1514)	All endpoint log data encrypted in transit
TLS 1.2/1.3	Layer 6	Wazuh Dashboard (HTTPS :443)	Browser ↔ Dashboard traffic
HTTPS	Layer 7	Wazuh API (:55000), S3 wodle polling	API management + CloudTrail log retrieval
SSH	Layer 7	Kali Linux port 22 — monitored by Rules 5710/5712	Secure remote access — brute force target
TLS (AWS)	Layer 6–7	All AWS API calls (CLI, SDK)	Cloud API traffic encryption

8. Project Results — Key Metrics

Metric	Target	Result
Attack detection rate	100%	✓ 100% — all 3 simulated attacks detected
SSH brute force detection	Rule 5712 fires	✓ Level 10 alert after 8 attempts / 120 seconds
FIM detection	All 50 files	✓ All 50 file events captured by inotify in real time
Cloud detection window	< 5 minutes	✓ Wazuh ingested CloudTrail logs within 5-min poll
Containment speed	< 60 seconds	✓ Shuffle SOAR isolated Kali endpoint in < 60s
MITRE coverage	3 attacks mapped	✓ TA0001 · TA0006 · TA0007 · TA0040 all documented
Compliance coverage	PCI/HIPAA/NIST	✓ PCI DSS 10.2.4 · HIPAA 164.312.b · NIST SI.4 · AU.14
Rubric completion	All items	✓ C1–C4 · R1–R5 · Network Security 1–7 all complete

The Sentinel Hybrid-SOC project successfully built, tested, and documented a fully functional Security Operations Center using open-source tooling on consumer hardware — demonstrating enterprise-grade detection and automated response capabilities across endpoint and cloud infrastructure.